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> **Note**: The factory image comes with the runtime environment preinstalled. **Warning**: The prediction mode of this CLI command reads video devices directly through OpenCV/V4L2. It primarily supports standard USB cameras and does not support the ROS driver node of the Nuwa depth camera.

1、Enable Optimal Board Performance

1.1、Enable Jetson Clocks

Enabling Jetson Clocks ensures that all CPU and GPU cores run at maximum frequency: **(Not required on RDK and Raspberry Pi boards)**

```
sudo jetson_clocks
```

2、Inference with the CLI

CLI (Command-Line Interface) is a way for users to interact with a computer or software program, in which programs are run or tasks are performed by typing text commands. Ultralytics provides a very convenient command-line interaction tool.

2.1、CLI Syntax

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track,
benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like
'imgsz=320' that override defaults.
```

2.2、Image Prediction

Use yolo26n.pt to run prediction on a target image: if the corresponding model file is not found in the directory where the command is executed, it will be downloaded automatically (if the network is unavailable, a local model can be used instead).

Navigate to the project folder:

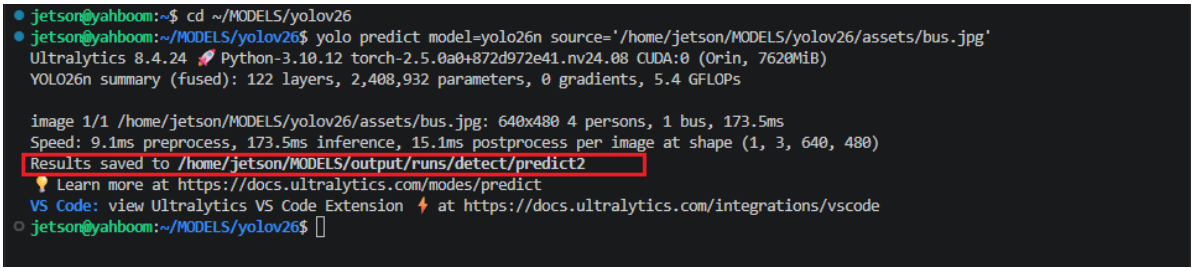
```
cd ~/MODELS/yolov26
```

Use yolo26n.pt to detect images in the target folder and output the results:

```
yolo predict model=yolo26n source='/home/jetson/MODELS/yolov26/assets/bus.jpg'

#rdk board
yolo predict model=yolo26n source='/home/sunrise/MODELS/yolov26/assets/bus.jpg'

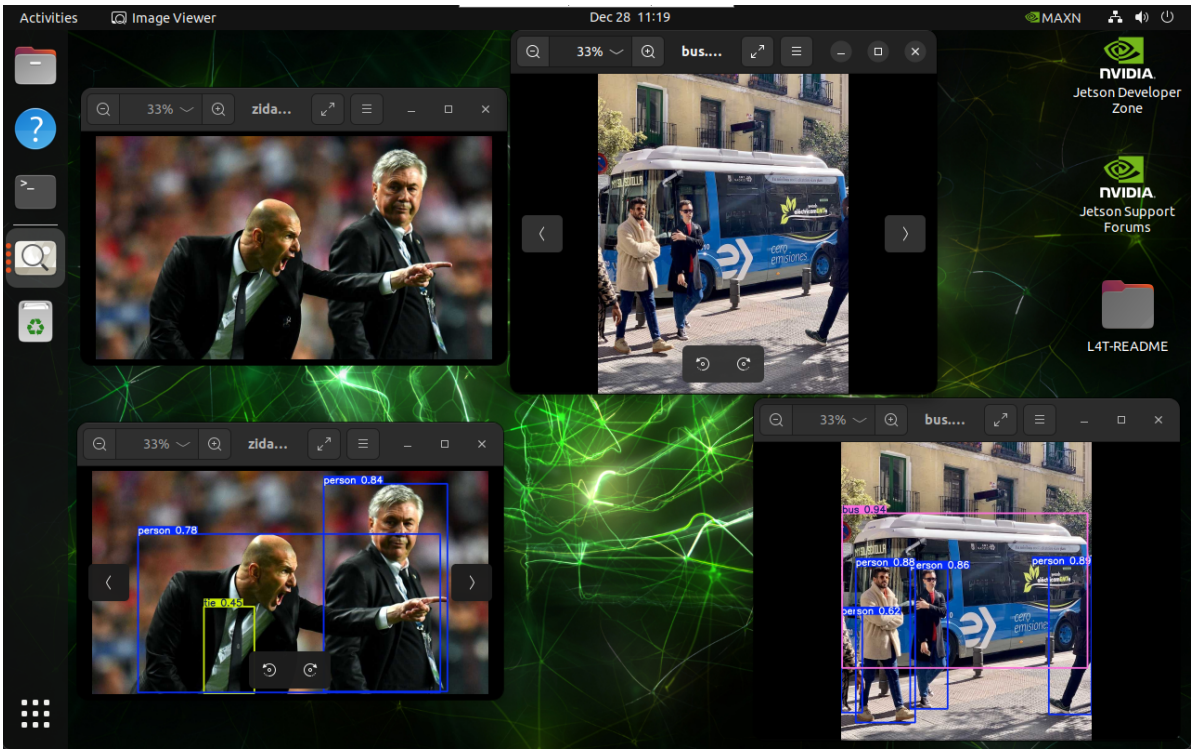
#Raspberry Pi board
yolo predict model=yolo26n source='/root/MODELS/yolov26/assets/bus.jpg'
```



Preview

Location of the images output by yolo: `/home/jetson/MODELS/output/runs/detect/predict*`
(On RDK and Raspberry Pi boards, refer to the actual path printed in the terminal)

With each run, the predict sequence number under this folder is incremented automatically:



2.3、Video Prediction

Use yolo26n.pt to run prediction on a target video file.

Navigate to the project folder:

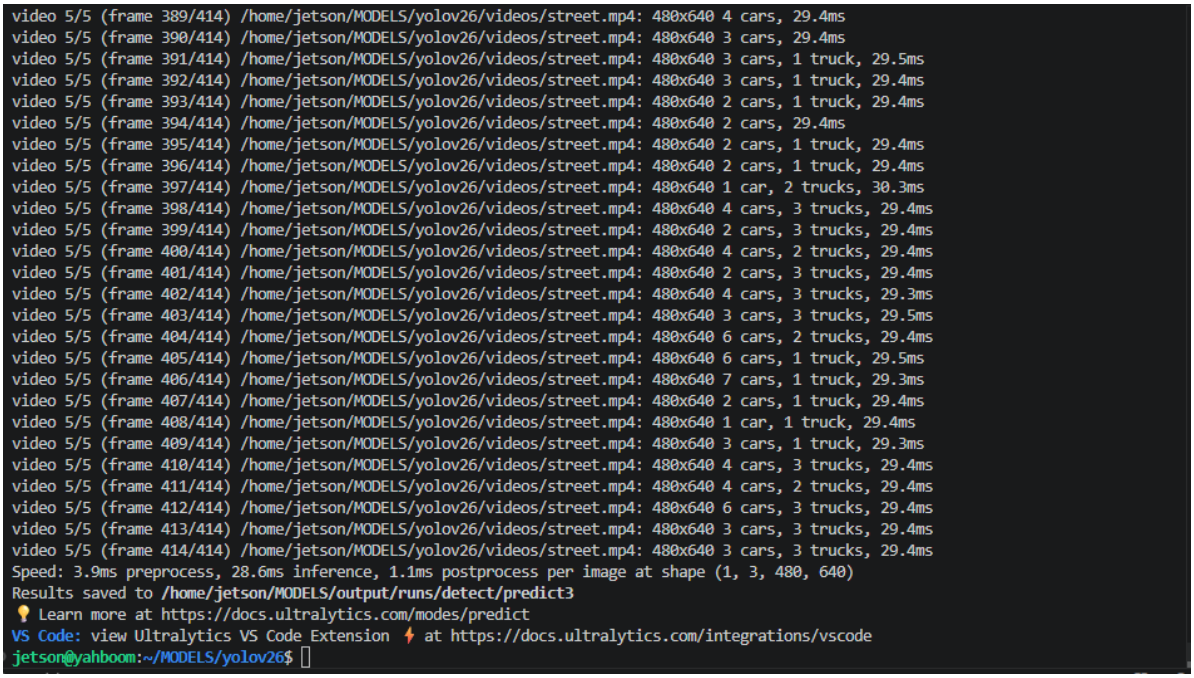
```
cd ~/MODELS/yolov26
```

Use yolo26n.pt to detect videos in the target folder and output the results:

```
yolo predict model=yolo26n source='/home/jetson/MODELS/yolov26/videos'

#rdk board
yolo predict model=yolo26n source='/home/sunrise/MODELS/yolov26/videos/car.mp4'

#Raspberry Pi board
yolo predict model=yolo26n source='/root/MODELS/yolov26/videos/car.mp4'
```



Preview

Location of the videos output by yolo: `/home/jetson/MODELS/output/runs/detect/predict*`
(On RDK and Raspberry Pi boards, refer to the actual path printed in the terminal)

With each run, the predict sequence number under this folder is incremented automatically. To view the generated video, use the ffmpeg/ffplay tools:

```
sudo apt update
sudo apt install ffmpeg -y
```

Play the video:

```
ffplay <video_name>.avi
```



2.4、Real-Time Prediction

Use yolo26n to run prediction on a live USB camera feed. **(This is not available for the depth camera because the video device driver cannot be used.)**

Navigate to the project folder:

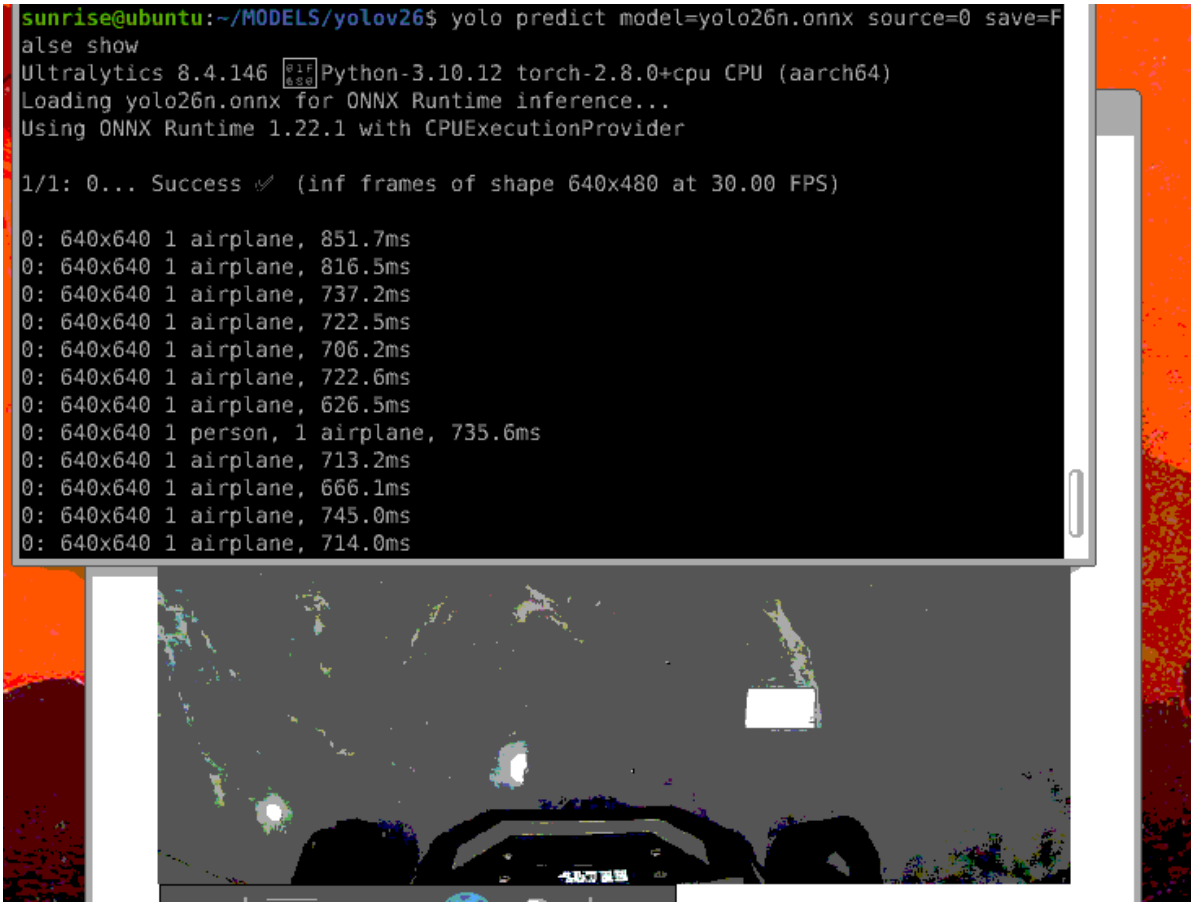
```
cd ~/MODELS/yolov26
```

Use yolo26n.pt to detect the camera feed and output the results:

```
yolo predict model=yolo26n.onnx source=0 save=False show # object detection
# instance segmentation: yolo predict model=yolo26n-seg.pt source=0 save=False
show
# image classification: yolo predict model=yolo26n-cls.pt source=0 save=False
show
# pose estimation: yolo predict model=yolo26n-pose.pt source=0 save=False show
# oriented object detection: yolo predict model=yolo26n-obb.pt source=0
save=False show
```

Click the terminal and press `Ctrl + C` to terminate the program.

> **Parameter Description**
>
> - `model`: Specifies the path or name of the YOLO model
> - `source`: Specifies the detection source (the USB camera device number is typically 0; change the number to switch between multiple cameras)
> - `save=False`: Disables saving result files to reduce storage usage
> - `show`: Displays the visualization window in real time



References

- Ultralytics repository: https://github.com/ultralytics/ultralytics
- CLI usage documentation: https://docs.ultralytics.com/usage/cli/